

# Exercise Sheet 2

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## Probability, Data and a Parameter

Probabilistic Machine Learning

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# 2.A Conditioning and Bayes

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# Exercise 2.A.1: Consequences of the Axioms

Using only the three Kolmogorov axioms (and set algebra), show:

1.  $P(A \setminus B) = P(A) - P(A \cap B)$
2.  $P(A \cup B) \leq P(A) + P(B)$
3. If  $A \subseteq B$ , then  $P(A) \leq P(B)$ .

# Exercise 2.A.1: Consequences of the Axioms — Solution

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1.  $A = (A \setminus B) \cup (A \cap B)$ , disjoint, so by Axiom III  $P(A) = P(A \setminus B) + P(A \cap B)$ .
2.  $A \cup B = A \cup (B \setminus A)$ , disjoint:  $P(A \cup B) = P(A) + P(B \setminus A) = P(A) + P(B) - P(A \cap B) \leq P(A) + P(B)$ , since  $P(A \cap B) \geq 0$  (Axiom I).
3.  $B = A \cup (B \setminus A)$ , disjoint:  $P(B) = P(A) + P(B \setminus A) \geq P(A)$  by Axioms III and I.

## Exercise 2.A.2: One Flag, Two Flags

Fraud occurs in **0.2%** of transactions. A rule flags **95%** of frauds and **2%** of legitimate transactions.

1. What fraction of flagged transactions is fraud?
2. A second, different rule has the same rates. Assume the two rules err **independently given the class**. A transaction is flagged by both. Now?
3. When would the assumption in 2 be badly wrong — and in which direction would the answer then be off?

# Exercise 2.A.2: One Flag, Two Flags — Solution

## Exercise 2.A.2: One Flag, Two Flags — Solution

1.  $P(\text{flag}) = 0.002 \cdot 0.95 + 0.998 \cdot 0.02 = 0.02186$ , so  $P(\text{fraud} \mid \text{flag}) = 0.0019/0.02186 \approx 8.7\%$ .
2. In odds: prior odds  $0.002/0.998 \approx 0.0020$ ; each flag multiplies them by the likelihood ratio  $0.95/0.02 = 47.5$ . Two flags:  $0.0020 \cdot 47.5^2 \approx 4.52$ , i.e.  $P \approx 4.52/5.52 \approx 82\%$ .
3. If both rules look at the same signal (say, both react to foreign IPs), the second flag adds little new evidence, yet the formula counts it twice — the answer is **too confident**. This is exactly the assumption naive Bayes makes.

## Exercise 2.A.3: Independent Is Not Disjoint

1. Show that two disjoint events  $A, B$  with  $P(A), P(B) > 0$  are never independent.
2. Flip two fair coins. Let  $A$  = "first is heads",  $B$  = "second is heads",  $C$  = "the coins agree". Show that  $A, B, C$  are pairwise independent but not mutually independent.

# Exercise 2.A.3: Independent Is Not Disjoint — Solution

## Exercise 2.A.3: Independent Is Not Disjoint — Solution

1.  $P(A \cap B) = 0 \neq P(A)P(B) > 0$ .
2. Each event has probability  $\frac{1}{2}$ .  $A \cap B = \{HH\}$ ,  $A \cap C = \{HH\}$ ,  $B \cap C = \{HH\}$ : each has probability  $\frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2}$  — pairwise independent. But  $A \cap B \cap C = \{HH\}$  has probability  $\frac{1}{4} \neq \frac{1}{8}$ . Knowing  $A$  and  $B$  determines  $C$ .

# 2.B Random Variables and the Joint

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## Exercise 2.B.1: App Use and Churn

The joint distribution of  $X$  (uses the app: 1/0) and  $Y$  (churns: 1/0):

|         | $Y = 0$ | $Y = 1$ |
|---------|---------|---------|
| $X = 0$ | 0.30    | 0.20    |
| $X = 1$ | 0.40    | 0.10    |

1. Compute  $P(X = 1)$ ,  $P(Y = 1)$ ,  $P(Y = 1 \mid X = 1)$  and  $P(Y = 1 \mid X = 0)$ .
2. Are  $X$  and  $Y$  independent? Compute  $\text{cov}(X, Y)$ .
3. Does the table show that the app *reduces* churn? What else could produce it?

# Exercise 2.B.1: App Use and Churn — Solution

## Exercise 2.B.1: App Use and Churn — Solution

1.  $P(X=1) = 0.5, P(Y=1) = 0.3, P(Y=1 \mid X=1) = 0.1/0.5 = 0.2, P(Y=1 \mid X=0) = 0.2/0.5 = 0.4$ .
2. Not independent ( $0.2 \neq 0.3$ ).  $\text{cov}(X, Y) = E[XY] - E[X]E[Y] = 0.10 - 0.5 \cdot 0.3 = -0.05$ .
3. No. A joint distribution shows association, not direction: loyal customers may be the ones who install the app (the abduction exercise of Lecture 01). Deciding needs an experiment or a model of the causes.

## Exercise 2.B.2: Dependent, Yet Uncorrelated

Let  $X$  take the values  $-1, 0, 1$  with probability  $\frac{1}{3}$  each, and  $Y = X^2$ .

1. Compute  $E[X]$ ,  $E[Y]$  and  $\text{cov}(X, Y)$ .
2. Show that  $X$  and  $Y$  are not independent.
3. What does this say about using correlation to screen features?

# Exercise 2.B.2: Dependent, Yet Uncorrelated — Solution

## Exercise 2.B.2: Dependent, Yet Uncorrelated — Solution

1.  $E[X] = 0, E[Y] = \frac{2}{3}, E[XY] = E[X^3] = 0$ , so  $\text{cov} = 0$ .
2.  $P(Y = 0 \mid X = 0) = 1 \neq P(Y = 0) = \frac{1}{3}$ .
3. A feature can determine the target completely and still have zero correlation with it. Correlation screens only for (monotone) linear association.

# 2.C Data, Statistics and a Parameter

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## Exercise 2.C.1: Scale and Model

For each attribute of a customer table, give the scale and a first probabilistic model, and name one statistic that would be **invalid**: payment method (card / invoice / PayPal) · delivery rating (1–5 stars) · number of orders last year · order value in € · has a newsletter subscription.

# Exercise 2.C.1: Scale and Model — Solution

# Exercise 2.C.1: Scale and Model — Solution

| attribute      | scale                   | first model                                 | invalid                                         |
|----------------|-------------------------|---------------------------------------------|-------------------------------------------------|
| payment method | nominal                 | categorical $(p_1, p_2, p_3)$               | mean, median                                    |
| rating         | ordinal                 | categorical (order unused) or ordinal model | mean (gaps unknown) — defensible only with care |
| orders         | ratio (count)           | Poisson, or binomial with a maximum         | — (all valid)                                   |
| order value    | ratio, positive, skewed | log-normal                                  | none, but the mean is outlier-sensitive         |
| newsletter     | nominal, binary         | Bernoulli( $p$ )                            | — (coded 0/1, the mean is $\hat{p}$ )           |

# Exercise 2.C.2: How Precise Is a Conversion Rate?

A campaign reached **400** users; **60** converted.

1. Estimate the conversion rate and its standard error.
2. Give an approximate 95% confidence interval and state correctly what the 95% refers to.
3. How many users would give a half-width of  $\pm 0.01$ ?

# Exercise 2.C.2: How Precise Is a Conversion Rate? — Solution

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1.  $\hat{p} = 60/400 = 0.15$ ;  $SE = \sqrt{0.15 \cdot 0.85/400} \approx 0.018$ .
2.  $0.15 \pm 1.96 \cdot 0.018 \approx [0.115, 0.185]$ . 95% of intervals built **by this procedure** contain the true rate — not "95% probability that the rate lies in this interval".
3.  $1.96\sqrt{0.1275/n} = 0.01 \Rightarrow n \approx 4,900$  — twelve times the data for a 3.5 times narrower interval:  
precision costs  $\sqrt{n}$ .

## Exercise 2.C.3: Two Analysts, One Campaign

Same data: 60 of 400 converted. Analyst A uses a flat prior  $\text{Beta}(1, 1)$ ; analyst B, from earlier campaigns, uses  $\text{Beta}(20, 80)$ .

1. Write down both posteriors and their means.
2. How many "pseudo-observations" is B's prior worth? Why do the two analysts nearly agree?
3. The data arrived in two batches: 25 of the first 200, 35 of the second 200. Show that updating twice gives the same posterior as updating once.
4. What is A's probability that the next user converts?

# Exercise 2.C.3: Two Analysts, One Campaign — Solution

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1. A:  $\text{Beta}(61, 341)$ , mean  $61/402 \approx 0.152$ . B:  $\text{Beta}(80, 420)$ , mean  $80/500 = 0.160$ . 95% credible intervals: A  $[0.118, 0.188]$ , B  $[0.129, 0.193]$ .
2.  $a + b = 100$  pseudo-observations against 400 real ones; B's prior mean 0.2 is pulled almost all the way to the data's 0.15.
3. A:  $\text{Beta}(1, 1) \rightarrow \text{Beta}(26, 176) \rightarrow \text{Beta}(61, 341)$  — counts simply add, in any order.
4. The posterior mean,  $\approx 0.152$ : averaging  $\theta$  over the posterior.

# Exercise 2.C.4: A Sampling Distribution by Simulation (Python)

```
import numpy as np
rng = np.random.default_rng(0)
samples = rng.exponential(scale=2.0, size=(1000, 25)) # 1000 samples, n = 25
```

1. Compute the mean of each sample and the standard deviation of those means. Compare with  $\sigma / \sqrt{n}$  (for this distribution  $\sigma = 2$ ).
2. The exponential is strongly skewed. Plot a histogram of the means — what shape do you see, and which theorem predicts it?
3. Repeat with  $n = 100$ . By what factor does the spread shrink?

# Exercise 2.C.4: A Sampling Distribution by Simulation — Solution

# Exercise 2.C.4: A Sampling Distribution by Simulation — Solution

```
means = samples.mean(axis=1)
print(means.std(ddof=1), 2.0 / np.sqrt(25)) # ≈ 0.40 vs 0.40
```

1. About 0.40 against  $\sigma/\sqrt{n} = 2/5 = 0.40$ .
2. Close to a bell, slightly right-skewed — the central limit theorem, not yet perfect at  $n = 25$  for a skewed population.
3. Four times the data, half the spread: about 0.20.